

Analysis of a Flawed Argument About Quantum Cloning

1 The Argument

Consider the following argument:

1. Any pure quantum state can be expressed in the $\{|0\rangle, |1\rangle\}$ basis.
2. The states in the $\{|0\rangle, |1\rangle\}$ basis can be perfectly copied.
3. Therefore, any quantum state can be perfectly copied.

At first glance, this reasoning seems plausible because quantum states can be expanded in terms of a basis. However, this argument is ****incorrect****. We will analyze why.

2 Step 1: Representing an arbitrary qubit

Let an arbitrary pure qubit state be

$$|\psi\rangle = \alpha |0\rangle + \beta |1\rangle, \quad \text{with } |\alpha|^2 + |\beta|^2 = 1. \quad (1)$$

It is true that $|\psi\rangle$ can be expressed in the $\{|0\rangle, |1\rangle\}$ basis.

3 Step 2: Cloning basis states

The argument claims that since $|0\rangle$ and $|1\rangle$ can be perfectly cloned, $|\psi\rangle$ can also be cloned.

Indeed, one can define a unitary U that clones basis states:

$$U |0\rangle \otimes |e\rangle = |0\rangle \otimes |0\rangle, \quad (2)$$

$$U |1\rangle \otimes |e\rangle = |1\rangle \otimes |1\rangle, \quad (3)$$

where $|e\rangle$ is the blank ancilla.

4 Step 3: Applying U to a superposition

Suppose we try to clone an arbitrary state $|\psi\rangle$ using U :

$$U |\psi\rangle \otimes |e\rangle = U(\alpha |0\rangle + \beta |1\rangle) \otimes |e\rangle \quad (4)$$

$$= \alpha U |0\rangle \otimes |e\rangle + \beta U |1\rangle \otimes |e\rangle \quad (\text{linearity of quantum mechanics}) \quad (5)$$

$$= \alpha |0\rangle \otimes |0\rangle + \beta |1\rangle \otimes |1\rangle. \quad (6)$$

Compare this with the *desired cloned state*:

$$|\psi\rangle \otimes |\psi\rangle = (\alpha |0\rangle + \beta |1\rangle) \otimes (\alpha |0\rangle + \beta |1\rangle) = \alpha^2 |0\rangle |0\rangle + \alpha\beta |0\rangle |1\rangle + \alpha\beta |1\rangle |0\rangle + \beta^2 |1\rangle |1\rangle. \quad (7)$$

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5 Step 4: The contradiction

The actual output of U is:

$$\alpha |0\rangle |0\rangle + \beta |1\rangle |1\rangle, \quad (8)$$

while the desired cloned state is:

$$\alpha^2 |0\rangle |0\rangle + \alpha\beta |0\rangle |1\rangle + \alpha\beta |1\rangle |0\rangle + \beta^2 |1\rangle |1\rangle. \quad (9)$$

These two expressions are **identical only if**:

$$\alpha\beta = 0, \quad (10)$$

i.e., only for the basis states themselves or when one of the coefficients is zero. For a **general superposition**, the two states are **not equal**.

This shows that a unitary operation that clones basis states cannot clone arbitrary superpositions.

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6 Step 5: The key lesson

The error in the argument is the **misinterpretation of linearity**:

- While any state can be expressed as a linear combination of basis states, quantum operations are linear.
- Cloning the basis states individually and then attempting to use linearity to extend to arbitrary superpositions fails.
- This is exactly why the **No-Cloning Theorem** holds: there is no universal unitary operator that can clone all unknown quantum states.

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7 Step 6: Conclusion

The argument fails because it **assumes that linearity allows perfect cloning of superpositions if basis states can be cloned**, which is false. Perfect cloning of arbitrary quantum states is impossible. Only orthogonal or known states can be cloned perfectly.

References:

1. W. K. Wootters and W. H. Zurek, “A single quantum cannot be cloned,” *Nature* 299, 802–803 (1982).
2. V. Bužek and M. Hillery, “Quantum copying: Beyond the no-cloning theorem,” *Phys. Rev. A* 54, 1844 (1996).
3. M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, 2010.